

CUBEATS: A Tangible 3D Step-Sequencer

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ABSTRACT

In an effort to break the mold of the standard two dimensional step-sequencer paradigm, a novel method of integrating a tangible third dimension or parameter of musical control is presented. Using clear plastic cubes and a grid of sensor detecting vertical rods, anyone from a child to an advanced electronic musician can build, manipulate, and perform musical patterns by stacking up to three cubes/rod in any position along a 4x16 grid. The vertical axis used not only provides additional control, but acts as a fun and easy way to engage the user and compose new musical patterns or sequences.

Keywords

Step sequencer, music interface, Hall Effect sensors, cubes, light-emitting diodes, tangible.

1. INTRODUCTION

Typical physical step sequencers operate on a flat two dimensional plane, providing only two control parameters within the potential three-dimensional space. The x-axis controls the time interval broken up into units corresponding to the common 4/4 time signature found in most sequenced electronic music. The y-axis provides voice or channel polyphony for multiple instrument control, while the z-axis is void of any interaction. For drum patterns or beats, the user can add a kick drum, snare drum, and a hi-hat on different voices for individual time control for each instrument, but cannot control the individual volume, alternate sample for the given sound or an alternate pitch using the analogous z-axis.

Using the vertical axis can resolve this issue of ill control not only for drum patterns but melodic patterns as well. For this work, clear stackable cubes were implemented with typical two-dimensional step sequencer control, resulting in a fun new interaction level for making sequenced music patterns. Up to three stackable cubes with drilled centers can be placed on a rigid vertical rod for each grid space of the sequencer. Once one or more cubes are placed on the rod, a single internal light-emitting diode (LED) is engaged and will flash on the given musical beat according to the set tempo and quantization scheme. The rods not

only guide the blocks in a stackable fashion, but improve the diffusion and appearance from the light below. The following will be a detailed explanation of the sensing scheme and overall design.

2. SENSING SCHEME

Designing a sensing scheme capable of the typical step sequencer behavior, while integrating vertical axis control, proved quite difficult. The one-inch plastic acrylic cubes, while priced at cost effective \$1 per unit, are very light-weight complicating the use of force sensitive resistors (FSR) and/or similar devices. Experimentally, a 25% variation of amplitude from a repeated force was found using a 1/2" circular FSR available from Interlink Electronics. Additionally, FSR sensors can cost up to \$8/unit, making a 64-grid design exceedingly expensive. To avoid using force sensors, the force or weight of the plastic cubes can be transferred into a displacement offset, making for a much easier sensing scheme.

By compressing a spring-like material with the weight of one or more cubes, displacement can be measured in numerous forms; initially, mechanical springs were used in numerous configurations with various sensors yielding frustrating results. After much deliberation, levitating magnets guided by clear plastic rods were implemented to capture the displacement compression using Hall Effect sensors (a magnetic field sensor). Priced at around \$1, Hall Effect sensors provide a cheap and relatively easy method for sensing the required vertical axis. Additionally, cheap generic square magnets were used (three/rod).

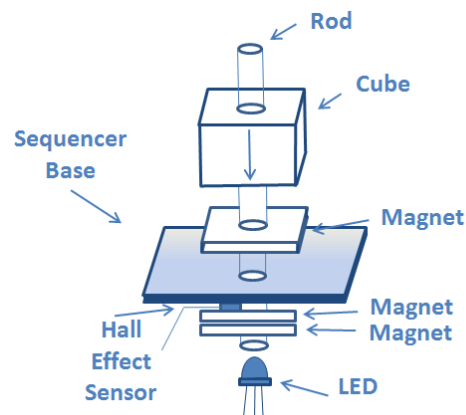


Figure 1. Sensor Scheme

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As seen in Fig. 1, two magnets are fastened to the underside of the sequencer—one directly attached to the sequencer base and one directly attached to the standing vertical rod. Opposite polarity fastens the rod to the sequencer base while simultaneously increasing the Hall Effect sensitivity and levitating force. The floating magnetic above (guided by the rod) is set in equilibrium nearly 2 cm above the base providing a respectable clearance to measure up to three individual cubes. As each cube is placed downward onto the floating magnet, a new equilibrium point is found closer to the base of the sequencer. This change in distance is measured via the Hall Effect sensor, which in turn outputs an increased analog voltage. This voltage level is sampled or discretized using an analog-to-digital converter onboard a microcontroller. For overall system design, see below.

3. OVERALL DESIGN

3.1 Hardware

Housed within a 20x16x6 inch Plexiglas black box, a total of 64 clear plastic rods, 64 Hall Effect sensors, 64 light-emitting diodes (LEDs), 1 Atmel AVR ATmega 32 microcontroller, and numerous multiplexers are used for the general design. All analog-to-digital (Hall Effect sensors) conversion is handled via the microcontroller with little extraneous circuitry.

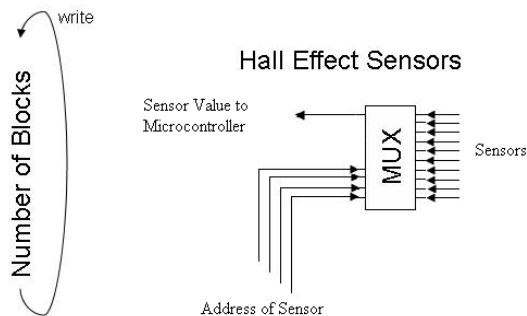


Figure 2. Simplified Multiplexer Scheme

As seen in Fig. 2, to handle the large input/output sensing scheme, however, numerous multiplexers are used to map multiple Hall Effect sensors onto a single analog-to-digital pin on the microcontroller (totaling four A/D pins for the 64 continuous sensors). Similarly, multiplexers are also used for the LED output control, sending a simple voltage on/off signal. As seen in Fig. 3, all electronic circuitry and components are then hidden inside the Plexiglas box.

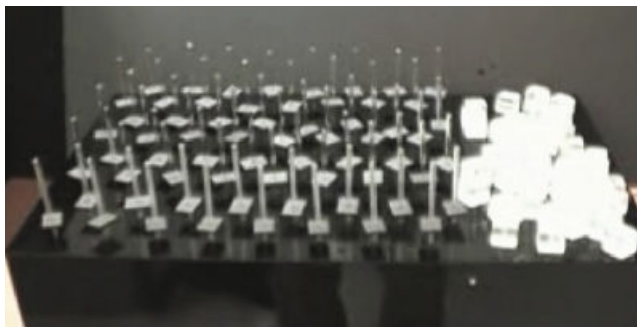


Figure 3. CUBEATS Step-Sequencer

3.2 Software

The microcontroller can easily connect to a personal computer via a Universal Serial Bus (USB) connection, programmed in the C programming language to input the sensing data, output the lighting data, and communicate to third-party audio applications using Open Sound Control (OSC) messages such as Pure Data (Pd) [1].

From the analog-to-digital conversion of the continuous sensor voltage levels, a maximum-likelihood threshold estimation is used to resolve the number of blocks/rod (i.e. 0, 1, 2, or 3). Large variability in sensors and magnet strength require an initial calibration phase, measuring the mean displacement per number of cubes. Once calibrated, the sequencer enters a ready state mode and begins to sample the sensor data and establish two-way communication to a personal computer.

The necessary information is then sent to Pd via OSC into a sequencer patch. The sequencer patch synchronizes the commands on light and sound to a beats per minute (BPM) rate and quantization ratio controlled by the user (the baseline quantization count is sent to the microcontroller at the respective tempo). Additionally, OSC message data rate is reduced for maximum performance by only sending changes in the number of block.

4. CONCLUSION

The breakout to three dimensions increases the typical interaction, engagement, and musical possibilities for tangible step-sequencers making CUBEATS a unique interface for sequenced musical patterns. Additionally, the Hall Effect/magnet design offers a relatively cheap solution for sensing large arrays of continuous data. Overall, the clear cubes, clear rods, black base, and colored LED feedback not only make the sequencer fun for individuals or small groups to play and interact with, but can also add a pleasant visual experience to a live music event or concert.

5. ACKNOWLEDGMENTS

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6. REFERENCES

- [1] *AVR-CCRM Wiki* [online]. Available: <http://cm-wiki.stanford.edu/wiki/AVR>